■ Routing Information Protocol (RIP)

■ 1. RIP (Overview)

• RIP = Distance Vector Routing Protocol.
• Uses Bellman–Ford Algorithm.
• Metric = hop count (max 15 hops).
• Protocol number: UDP port 520.
• Designed for small to medium networks.

■ 2. RIP Updating Algorithm

Core formula (Bellman–Ford based):

$$D_x(y) = \min_{v \in N(x)} \left(c(x,v) + D_v(y) \right)$$

Where:
• $D_x(y)$ = cost from router x → destination y
• $c(x,v)$ = cost of link (x,v)
• $D_v(y)$ = cost from neighbor v → y

■ Each router sends its entire routing table to neighbors every 30 seconds.
■ Neighbors update their table with the minimum cost path.

■ 3. RIP Operations

1. Initialization – Router sets distance to itself = 0, others = ∞.
2. Request – Router can send a request for routing tables.
3. Response – Router replies with routing table entries.
4. Periodic Update – Every 30 sec, send full table to neighbors.
5. Triggered Update – If change detected (e.g., link fails), send immediate update.

■ 4. RIP Message Format

RIP messages are carried in UDP datagrams.

Fields in RIP Packet:
• Command (1 byte) → 1=Request, 2=Response
• Version (1 byte) → RIP v1=1, RIP v2=2
• Unused (2 bytes) → must be 0
• Entries (20 bytes each):
  - Address family identifier (AFI)
  - IP Address
  - Metric (hop count: 1–15; 16 = infinity)

■ 5. Request and Response Messages

• Request Message → Router asks neighbor to send routing table.
• Response Message → Contains routing table (either full table or requested part).
• Sent as:

- Periodic updates (every 30s).
- Triggered updates (on change).

■ 6. Timers in RIP

1. Update Timer (30 sec) → send routing table.
2. Invalid Timer (180 sec) → if no update, mark route invalid.
3. Hold-down Timer (180 sec) → prevents flapping; ignores bad info until timer expires.
4. Flush Timer (240 sec) → remove route from table if still invalid.

■ 7. Disadvantages of RIPv1

• Only supports classful routing (no subnet info).
• No authentication (prone to attacks).
• Maximum hop count = 15 (limits scalability).
• Slow convergence.
• Count-to-Infinity problem.

■ 8. RIP Version 2 (RIPv2) Features

• Supports classless routing (subnet masks included).
• Multicasts updates to 224.0.0.9 (instead of broadcast).
• Supports authentication.
• Route Tag field (useful in policy routing).
• Backward compatible with RIPv1.

■ 9. Authentication in RIP

• Plaintext Authentication → password included (not secure).
• MD5 Authentication → hash-based, secure.
• Ensures only trusted routers exchange updates.

■ 10. Problems in RIP

• Count-to-Infinity Problem → solved with Split Horizon, Poison Reverse.
• Scalability Issues → hop limit of 15 restricts large networks.
• Slow Convergence → takes time to adapt after topology changes.
• Vulnerable to routing table poisoning without authentication.